University of Dhaka

Department of Computer Science and Engineering, Faculty of Engineering & Technology

CSE 2205: Microcontroller and Embedded System

2nd Year, 2nd Semester | 2024

QUIZ 1
A
SET

Quiz	1
Total Marks	30
Marks per Q	1 mark
Negative marking	None
Duration	40 minutes
Answer sheet	OMR (HB/2B pencil)

OBE Alignment

Course Outcome	Bloom	Qs
CO1 – MCU architecture (PO1)	C3	12
CO2 – MCU operations (PO2)	C4	18
Total		**30**

Instructions: (**1**) 30 questions; 1 mark each; **no negative marking.** (**2**) All sets (A, B, C) share the same questions in different order with shuffled choices; answer **only your assigned set.** (**3**) Mark answers on the **OMR sheet** — HB or 2B pencil only. (**4**) No mobile phones or electronic devices.

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1. A GPIO pin has MODER = `10`. Which mode is selected?
(A) Input (reset state) **(B)** Alternate function **(C)** Analog **(D)** General purpose output

2. To enable HSE and wait for stability, which RCC_CR bits are used in sequence?
(A) Set PLLON, poll PLLRDY **(B)** Set HSEON, poll HSERDY **(C)** Set HSION, poll HSIRDY **(D)** Set CSSON, poll HSERDY

3. How many bits per pin does GPIOx_MODER use to configure the pin mode?
(A) 1 bit **(B)** 2 bits **(C)** 8 bits **(D)** 4 bits

4. In PWM generation, duty cycle is primarily set by:
(A) TIMx_CNT **(B)** TIMx_ARR **(C)** TIMx_CCRx **(D)** TIMx_PSC

5. For USART2 at 115200 baud with APB1 = 45 MHz, the USART_BRR mantissa is:
(A) 156 **(B)** 390 **(C)** 781 **(D)** 234

6. In the STM32F446RE memory map, the base address of GPIOA is:
(A) 0x4002_3800 **(B)** 0x4000_0000 **(C)** 0x4002_0000 **(D)** 0x4001_0000

7. The NVIC in STM32F446RE supports how many configurable interrupt priority levels?
(A) 8 **(B)** 32 **(C)** 16 **(D)** 4

8. The PLL system core clock formula is:
(A) $\text{SysClk} = \dfrac{(\text{IN}/\text{PLLM}) \times \text{PLLN}}{\text{PLLP}}$ **(B)** $\text{SysClk} = \dfrac{\text{IN} \times \text{PLLM}}{\text{PLLN} \times \text{PLLP}}$ **(C)** $\text{SysClk} = \text{IN} \times \text{PLLM} \times \text{PLLN}$ **(D)** $\text{SysClk} = \dfrac{\text{IN}}{\text{PLLM} + \text{PLLN} + \text{PLLP}}$

9. In I^2C, a HIGH-to-LOW transition on SDA while SCL is HIGH defines:
(A) NACK condition **(B)** START condition **(C)** ACK condition **(D)** STOP condition

10. After PLL setup, the correct final step to switch SYSCLK to the PLL is:
(A) Reset the MCU so the PLL activates on boot **(B)** Write SW = `10` in RCC_CFGR, then poll SWS = `10` **(C)** Modify PLLSRC in RCC_PLLCFGR directly **(D)** Set PLLON and read SWS bits

11. Which GPIO output mode can actively drive the pin both HIGH and LOW using complementary MOSFETs?
(A) Alternate function input **(B)** Analog input **(C)** Push-pull **(D)** Open-drain

12. USARTx_CR2 bits [13:12] = `2'b10` selects:
(A) 2 stop bits **(B)** 0.5 stop bit **(C)** 1.5 stop bits **(D)** 1 stop bit

13. I^2C master, standard mode (100 kHz SCL), APB1 = 45 MHz. CCR ≈:
(A) 450 **(B)** 225 **(C)** 45 **(D)** 100

14. For a 1 ms tick with TIM6 at APB1 timer clock 90 MHz, PSC should be:
(A) 179999 **(B)** 8999 **(C)** 89999 **(D)** 44999

15. For 180 MHz operation, how many wait-states must be set in FLASH_ACR?
(A) 5 WS **(B)** 7 WS **(C)** 2 WS **(D)** 3 WS

16. TIM1: ARR = 999, PSC = 179, APB2 timer clk = 180 MHz. PWM frequency:
(A) 100 kHz **(B)** 500 Hz **(C)** 10 kHz **(D)** 1 kHz

17. What is the maximum system clock frequency of the STM32F446RE?
(A) 180 MHz **(B)** 168 MHz **(C)** 120 MHz **(D)** 72 MHz

18. In RCC_PLLCFGR, what division factor does PLLP = `2'b00` select?
(A) 4 **(B)** 8 **(C)** 1 **(D)** 2

19. In GPIOx_BSRR, which bits RESET (clear to logic 0) an output pin?
(A) Bits 8–15 **(B)** All 32 bits **(C)** Bits 0–15 **(D)** Bits 16–31

20. Which bus in STM32F446RE carries GPIO ports A through H?
(A) APB1 **(B)** AHB3 **(C)** AHB1 **(D)** APB2

21. To configure a GPIO pin for USART alternate function, which register is written?
(A) GPIOx_AFR[0]/AFR[1] **(B)** GPIOx_IDR **(C)** GPIOx_ODR **(D)** GPIOx_BSRR

22. The duty cycle D of a PWM signal is:
(A) $D = \dfrac{t_{\text{on}}}{t_{\text{on}} + t_{\text{off}}}$ **(B)** $D = \dfrac{t_{\text{on}} - t_{\text{off}}}{t_{\text{on}} + t_{\text{off}}}$ **(C)** $D = t_{\text{on}} \times f$ **(D)** $D = \dfrac{t_{\text{off}}}{t_{\text{on}} + t_{\text{off}}}$

23. The Cortex-M4 pipeline in STM32F446RE consists of how many stages?
(A) 5: Fetch→Decode→Issue→Execute→WB **(B)** 4: Fetch→Decode→Execute→Memory **(C)** 2: Fetch→Execute **(D)** 3: Fetch→Decode→Execute

24. In USARTx_SR, which flag means the TX data register is empty and ready?
(A) IDLE **(B)** TC **(C)** RXNE **(D)** TXE

25. Which of the following is NOT a valid clock source for the STM32F446RE PLL?
(A) HSE (e.g., 8 MHz crystal) **(B)** LSE (32.768 kHz RTC crystal) **(C)** Both HSI and HSE can feed the PLL **(D)** HSI (16 MHz internal RC)

26. Which timers in STM32F446RE are *Basic Timers* used primarily to drive the DAC?
(A) TIM6 & TIM7 **(B)** TIM2 & TIM5 **(C)** TIM1 & TIM8 **(D)** TIM9 & TIM12

27. In USART setup, which step must come **first**?
(A) Enable the USART peripheral clock in RCC_APBxENR **(B)** Enable UE in USARTx_CR1 **(C)** Set TE and RE in USARTx_CR1 **(D)** Set baud rate in USART_BRR

28. The STM32F446RE microcontroller is based on which processor core?
(A) ARM Cortex-M7 with double-precision FPU **(B)** ARM Cortex-M0 with no FPU **(C)** ARM Cortex-M3 **(D)** ARM Cortex-M4 with FPU

29. The APB1 peripheral bus in STM32F446RE has a maximum frequency of:
(A) 180 MHz **(B)** 45 MHz **(C)** 90 MHz **(D)** 22.5 MHz

30. Which bit starts the counter in TIM6 or TIM7?
(A) ARPE in TIMx_CR1 **(B)** UIE in TIMx_DIER **(C)** CEN in TIMx_CR1 **(D)** URS in TIMx_CR1

Quiz	1
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Negative marking	None
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OBE Alignment

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CO1 – MCU architecture (PO1)	C3	12
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Instructions: (**1**) 30 questions; 1 mark each; **no negative marking.** (**2**) All sets (A, B, C) share the same questions in different order with shuffled choices; answer **only your assigned set.** (**3**) Mark answers on the **OMR sheet** — HB or 2B pencil only. (**4**) No mobile phones or electronic devices.

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1. To enable HSE and wait for stability, which RCC_CR bits are used in sequence?
(A) Set CSSON, poll HSERDY **(B)** Set HSEON, poll HSERDY **(C)** Set PLLON, poll PLLRDY **(D)** Set HSION, poll HSIRDY

2. In USART setup, which step must come **first**?
(A) Set TE and RE in USARTx_CR1 **(B)** Enable the USART peripheral clock in RCC_APBxENR **(C)** Enable UE in USARTx_CR1 **(D)** Set baud rate in USART_BRR

3. After PLL setup, the correct final step to switch SYSCLK to the PLL is:
(A) Modify PLLSRC in RCC_PLLCFGR directly **(B)** Write SW = `10` in RCC_CFGR, then poll SWS = `10` **(C)** Reset the MCU so the PLL activates on boot **(D)** Set PLLON and read SWS bits

4. In I^2C, a HIGH-to-LOW transition on SDA while SCL is HIGH defines:
(A) ACK condition **(B)** NACK condition **(C)** STOP condition **(D)** START condition

5. For 180 MHz operation, how many wait-states must be set in FLASH_ACR?
(A) 2 WS **(B)** 7 WS **(C)** 3 WS **(D)** 5 WS

6. Which bit starts the counter in TIM6 or TIM7?
(A) UIE in TIMx_DIER **(B)** URS in TIMx_CR1 **(C)** CEN in TIMx_CR1 **(D)** ARPE in TIMx_CR1

7. To configure a GPIO pin for USART alternate function, which register is written?
(A) GPIOx_ODR **(B)** GPIOx_BSRR **(C)** GPIOx_AFR[0]/AFR[1] **(D)** GPIOx_IDR

8. Which of the following is NOT a valid clock source for the STM32F446RE PLL?
(A) HSI (16 MHz internal RC) **(B)** HSE (e.g., 8 MHz crystal) **(C)** Both HSI and HSE can feed the PLL **(D)** LSE (32.768 kHz RTC crystal)

9. In PWM generation, duty cycle is primarily set by:
(A) TIMx_CCRx **(B)** TIMx_CNT **(C)** TIMx_PSC **(D)** TIMx_ARR

10. A GPIO pin has MODER = `10`. Which mode is selected?
(A) General purpose output **(B)** Analog **(C)** Input (reset state) **(D)** Alternate function

11. Which GPIO output mode can actively drive the pin both HIGH and LOW using complementary MOSFETs?
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12. The PLL system core clock formula is:
(A) $\text{SysClk} = \dfrac{\text{IN} \times \text{PLLM}}{\text{PLLN} \times \text{PLLP}}$ **(B)** $\text{SysClk} = \dfrac{\text{IN}}{\text{PLLM} + \text{PLLN} + \text{PLLP}}$ **(C)** $\text{SysClk} = \dfrac{(\text{IN}/\text{PLLM}) \times \text{PLLN}}{\text{PLLP}}$ **(D)** $\text{SysClk} = \text{IN} \times \text{PLLM} \times \text{PLLN}$

13. TIM1: ARR = 999, PSC = 179, APB2 timer clk = 180 MHz. PWM frequency:
(A) 1 kHz **(B)** 500 Hz **(C)** 10 kHz **(D)** 100 kHz

14. What is the maximum system clock frequency of the STM32F446RE?
(A) 180 MHz **(B)** 168 MHz **(C)** 120 MHz **(D)** 72 MHz

15. I^2C master, standard mode (100 kHz SCL), APB1 = 45 MHz. CCR ≈:
(A) 45 **(B)** 225 **(C)** 100 **(D)** 450

16. Which bus in STM32F446RE carries GPIO ports A through H?
(A) APB2 **(B)** AHB3 **(C)** AHB1 **(D)** APB1

17. The APB1 peripheral bus in STM32F446RE has a maximum frequency of:
(A) 22.5 MHz **(B)** 45 MHz **(C)** 180 MHz **(D)** 90 MHz

18. For a 1 ms tick with TIM6 at APB1 timer clock 90 MHz, PSC should be:
(A) 89999 **(B)** 179999 **(C)** 44999 **(D)** 8999

19. In the STM32F446RE memory map, the base address of GPIOA is:
(A) 0x4000_0000 **(B)** 0x4002_3800 **(C)** 0x4002_0000 **(D)** 0x4001_0000

20. The Cortex-M4 pipeline in STM32F446RE consists of how many stages?
(A) 3: Fetch→Decode→Execute **(B)** 4: Fetch→Decode→Execute→Memory **(C)** 2: Fetch→Execute **(D)** 5: Fetch→Decode→Issue→Execute→WB

21. The duty cycle D of a PWM signal is:
(A) $D = \frac{t_{\text{on}}}{t_{\text{on}} + t_{\text{off}}}$ **(B)** $D = t_{\text{on}} \times f$ **(C)** $D = \frac{t_{\text{on}} - t_{\text{off}}}{t_{\text{on}} + t_{\text{off}}}$ **(D)** $D = \frac{t_{\text{off}}}{t_{\text{on}} + t_{\text{off}}}$

22. How many bits per pin does GPIOx_MODER use to configure the pin mode?
(A) 1 bit **(B)** 8 bits **(C)** 2 bits **(D)** 4 bits

23. In RCC_PLLCFGR, what division factor does PLLP = `2'b00` select?
(A) 2 **(B)** 4 **(C)** 1 **(D)** 8

24. For USART2 at 115200 baud with APB1 = 45 MHz, the USART_BRR mantissa is:
(A) 390 **(B)** 781 **(C)** 156 **(D)** 234

25. The NVIC in STM32F446RE supports how many configurable interrupt priority levels?
(A) 32 **(B)** 16 **(C)** 8 **(D)** 4

26. In USARTx_SR, which flag means the TX data register is empty and ready?
(A) TC **(B)** TXE **(C)** RXNE **(D)** IDLE

27. The STM32F446RE microcontroller is based on which processor core?
(A) ARM Cortex-M0 with no FPU **(B)** ARM Cortex-M4 with FPU **(C)** ARM Cortex-M3 **(D)** ARM Cortex-M7 with double-precision FPU

28. USARTx_CR2 bits [13:12] = `2'b10` selects:
(A) 2 stop bits **(B)** 1.5 stop bits **(C)** 1 stop bit **(D)** 0.5 stop bit

29. Which timers in STM32F446RE are *Basic Timers* used primarily to drive the DAC?
(A) TIM2 & TIM5 **(B)** TIM1 & TIM8 **(C)** TIM9 & TIM12 **(D)** TIM6 & TIM7

30. In GPIOx_BSRR, which bits RESET (clear to logic 0) an output pin?
(A) Bits 0–15 **(B)** All 32 bits **(C)** Bits 8–15 **(D)** Bits 16–31

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QUIZ 1

C

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OBE Alignment

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1. In RCC_PLLCFGR, what division factor does PLLP = `2'b00` select?
(A) 2 **(B)** 1 **(C)** 4 **(D)** 8

2. To enable HSE and wait for stability, which RCC_CR bits are used in sequence?
(A) Set HSEON, poll HSERDY **(B)** Set HSION, poll HSIRDY **(C)** Set CSSON, poll HSERDY **(D)** Set PLLON, poll PLLRDY

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6. After PLL setup, the correct final step to switch SYSCLK to the PLL is:
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7. Which timers in STM32F446RE are *Basic Timers* used primarily to drive the DAC?
(A) TIM9 & TIM12 **(B)** TIM6 & TIM7 **(C)** TIM2 & TIM5 **(D)** TIM1 & TIM8

8. For 180 MHz operation, how many wait-states must be set in FLASH_ACR?
(A) 7 WS **(B)** 3 WS **(C)** 5 WS **(D)** 2 WS

9. TIM1: ARR = 999, PSC = 179, APB2 timer clk = 180 MHz. PWM frequency:
(A) 1 kHz **(B)** 10 kHz **(C)** 100 kHz **(D)** 500 Hz

10. I^2C master, standard mode (100 kHz SCL), APB1 = 45 MHz. CCR ≈:
(A) 45 **(B)** 100 **(C)** 225 **(D)** 450

11. USARTx_CR2 bits [13:12] = `2'b10` selects:
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12. The NVIC in STM32F446RE supports how many configurable interrupt priority levels?
(A) 8 **(B)** 4 **(C)** 16 **(D)** 32

13. How many bits per pin does GPIOx_MODER use to configure the pin mode?
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(A) 3: Fetch→Decode→Execute **(B)** 4: Fetch→Decode→Execute→Memory **(C)** 5: Fetch→Decode→Issue→Execute→
(D) 2: Fetch→Execute

16. The PLL system core clock formula is:
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17. The APB1 peripheral bus in STM32F446RE has a maximum frequency of:
(A) 45 MHz **(B)** 180 MHz **(C)** 90 MHz **(D)** 22.5 MHz

18. The duty cycle D of a PWM signal is:
(A) $D = t_{\text{on}} \times f$ **(B)** $D = \dfrac{t_{\text{on}} - t_{\text{off}}}{t_{\text{on}} + t_{\text{off}}}$ **(C)** $D = \dfrac{t_{\text{off}}}{t_{\text{on}} + t_{\text{off}}}$ **(D)** $D = \dfrac{t_{\text{on}}}{t_{\text{on}} + t_{\text{off}}}$

19. In USART setup, which step must come **first**?
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(A) Alternate function input **(B)** Open-drain **(C)** Push-pull **(D)** Analog input

21. Which of the following is NOT a valid clock source for the STM32F446RE PLL?
(A) HSI (16 MHz internal RC) **(B)** HSE (e.g., 8 MHz crystal) **(C)** LSE (32.768 kHz RTC crystal) **(D)** Both HSI and HSE can feed the PLL

22. What is the maximum system clock frequency of the STM32F446RE?
(A) 180 MHz **(B)** 72 MHz **(C)** 168 MHz **(D)** 120 MHz

23. Which bus in STM32F446RE carries GPIO ports A through H?
(A) APB1 **(B)** APB2 **(C)** AHB1 **(D)** AHB3

24. For USART2 at 115200 baud with APB1 = 45 MHz, the USART_BRR mantissa is:
(A) 390 **(B)** 156 **(C)** 781 **(D)** 234

25. In the STM32F446RE memory map, the base address of GPIOA is:
(A) 0x4001_0000 **(B)** 0x4002_3800 **(C)** 0x4002_0000 **(D)** 0x4000_0000

26. In GPIOx_BSRR, which bits RESET (clear to logic 0) an output pin?
(A) Bits 0–15 **(B)** All 32 bits **(C)** Bits 8–15 **(D)** Bits 16–31

27. In I^2C, a HIGH-to-LOW transition on SDA while SCL is HIGH defines:
(A) NACK condition **(B)** ACK condition **(C)** START condition **(D)** STOP condition

28. A GPIO pin has MODER = `10`. Which mode is selected?
(A) Input (reset state) **(B)** General purpose output **(C)** Analog **(D)** Alternate function

29. Which bit starts the counter in TIM6 or TIM7?
(A) CEN in TIMx_CR1 **(B)** ARPE in TIMx_CR1 **(C)** URS in TIMx_CR1 **(D)** UIE in TIMx_DIER

30. In PWM generation, duty cycle is primarily set by:
(A) TIMx_CNT **(B)** TIMx_PSC **(C)** TIMx_ARR **(D)** TIMx_CCRx

Answer Key – Instructor Copy Only

Same 30 questions in all sets; question order and choice order differ between sets.

SET A

Q	Ans	CO	Bl.	Topic
1	**B**	CO2	C4	GPIO
2	**B**	CO2	C4	Clock
3	**B**	CO1	C3	GPIO
4	**C**	CO2	C4	PWM
5	**B**	CO2	C4	USART
6	**C**	CO1	C3	Memory
7	**C**	CO1	C3	Interrupt
8	**A**	CO2	C4	Clock
9	**B**	CO1	C3	I2C
10	**B**	CO2	C4	Clock
11	**C**	CO1	C3	GPIO
12	**A**	CO2	C4	USART
13	**B**	CO2	C4	I2C
14	**C**	CO2	C4	Timer
15	**A**	CO2	C4	Memory
16	**D**	CO2	C4	PWM
17	**A**	CO1	C3	Architecture
18	**D**	CO2	C4	Clock
19	**D**	CO2	C4	GPIO
20	**C**	CO1	C3	Architecture
21	**A**	CO2	C4	GPIO
22	**A**	CO2	C4	PWM
23	**D**	CO1	C3	Architecture
24	**D**	CO2	C4	USART
25	**B**	CO1	C3	Clock
26	**A**	CO1	C3	Timer
27	**A**	CO2	C4	USART
28	**D**	CO1	C3	Architecture
29	**B**	CO1	C3	Architecture
30	**C**	CO2	C4	Timer

Answer string: `BBBCB CCABB CABCA DADDC AADDB`

SET B

Q	Ans	CO	Bl.	Topic
1	**B**	CO2	C4	Clock
2	**B**	CO2	C4	USART
3	**B**	CO2	C4	Clock
4	**D**	CO1	C3	I2C
5	**D**	CO2	C4	Memory
6	**C**	CO2	C4	Timer
7	**C**	CO2	C4	GPIO
8	**D**	CO1	C3	Clock
9	**A**	CO2	C4	PWM
10	**D**	CO2	C4	GPIO
11	**B**	CO1	C3	GPIO
12	**C**	CO2	C4	Clock
13	**A**	CO2	C4	PWM
14	**A**	CO1	C3	Architecture
15	**B**	CO2	C4	I2C
16	**C**	CO1	C3	Architecture
17	**B**	CO1	C3	Architecture
18	**A**	CO2	C4	Timer
19	**C**	CO1	C3	Memory
20	**A**	CO1	C3	Architecture
21	**A**	CO2	C4	PWM
22	**C**	CO1	C3	GPIO
23	**A**	CO2	C4	Clock
24	**A**	CO2	C4	USART
25	**B**	CO1	C3	Interrupt
26	**B**	CO2	C4	USART
27	**B**	CO1	C3	Architecture
28	**A**	CO2	C4	USART
29	**D**	CO1	C3	Timer
30	**D**	CO2	C4	GPIO

Answer string: `BBBDD CCDAD BCAAB CBACA ACAAB`

SET C

Q	Ans	CO	Bl.	Topic
1	**A**	CO2	C4	Clock
2	**A**	CO2	C4	Clock
3	**B**	CO2	C4	USART
4	**A**	CO2	C4	Timer
5	**B**	CO1	C3	Architecture
6	**D**	CO2	C4	Clock
7	**B**	CO1	C3	Timer
8	**C**	CO2	C4	Memory
9	**A**	CO2	C4	PWM
10	**C**	CO2	C4	I2C
11	**B**	CO2	C4	USART
12	**C**	CO1	C3	Interrupt
13	**C**	CO1	C3	GPIO
14	**C**	CO2	C4	GPIO
15	**A**	CO1	C3	Architecture
16	**B**	CO2	C4	Clock
17	**A**	CO1	C3	Architecture
18	**D**	CO2	C4	PWM
19	**A**	CO2	C4	USART
20	**C**	CO1	C3	GPIO
21	**C**	CO1	C3	Clock
22	**A**	CO1	C3	Architecture
23	**C**	CO1	C3	Architecture
24	**A**	CO2	C4	USART
25	**C**	CO1	C3	Memory
26	**D**	CO2	C4	GPIO
27	**C**	CO1	C3	I2C
28	**D**	CO2	C4	GPIO
29	**A**	CO2	C4	Timer
30	**D**	CO2	C4	PWM

Answer string: `AABAB DBCAC BCCCA BADAC CACAC DCDAD`

Worked Solutions:

- **PLL SysClk:** SysClk $= \frac{(8/4)\times 180}{2} = 180\,\text{MHz}$ [HSE=8 MHz, PLLM=4, PLLN=180, PLLP=2].
- **Flash WS at 180 MHz:** 5 wait-states (STM32F446 DS, VDD=3.3 V).
- **USART2 BRR:** $45\text{M}/115200 \approx 390.625$; Mantissa=390, Fraction=10.
- **TIM6 PSC (1 ms):** APB1 tmr clk=90 MHz; PSC=$90\text{M}/1000 - 1 = 89999$.
- **PWM f (TIM1):** $f = 180\text{M}/[(179+1)(999+1)] = 1000\,\text{Hz}$=1 kHz.
- **I^2C CCR:** $(4000+1000)\,\text{ns} \div (1/45\,\text{MHz}) \approx 225$.